

SC7 Bayes Methods MT25

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Lecture 15: Bayesian non-parametrics : the CRP and inference

Notation... repeated values

Simulating $\theta_i \sim G$, $i = 1, \dots, n$ may give repeated θ -values (if we choose one of the “old” θ 's). Write

$$\sum_{i=1}^n \delta_{\theta_i} = \sum_{k=1}^K n_k \delta_{\theta_k^*}.$$

$\theta^* = (\theta_1^*, \dots, \theta_K^*)$ are the unique θ -values in $(\theta_1, \dots, \theta_n)$.

$K \geq 1$ is random, the number of distinct values in $(\theta_1, \dots, \theta_n)$.

n_k is the number of times θ_k^* appears in $(\theta_1, \dots, \theta_n)$.

For $k = 1, \dots, K$ let $S_k = \{i; \theta_i = \theta_k^*\}$ so we have a partition $S = \{S_1, \dots, S_K\}$ with $n_k = |S_k|$.

The mapping $\theta \rightarrow (\theta^*, S)$ is invertible: for $i = 1, \dots, n$ let $k_i = \{k : i \in S_k\}$; since S is a partition k_i is unique; set $\theta_i = \theta_{k_i}^*$.

Blackwell-MacQueen Urn Scheme

Sequential simulation for θ^*, S :

1. $\theta_1^* \sim H$; set $K = 1$, $S_1 = 1$ and $S = \{S_1\}$.
2. for $j = 1, \dots, n - 1$
 - (a) With probability $\alpha/(\alpha + j)$: simulate $\theta_{K+1}^* \sim H$; set $S_{K+1} = \{j + 1\}$ and $S \leftarrow S \cup \{S_{K+1}\}$; set $K \leftarrow K + 1$.
 - (b) Otherwise: for $k = 1, \dots, K$ set $n_k = |S_k|$; simulate $k \sim (n_1, \dots, n_K)/j$; set $S_k \leftarrow S_k \cup \{j + 1\}$.
3. For $j = 1, \dots, n$ set $k_j = \{k \in [K] : j \in S_k\}$ and $\theta_j = \theta_{k_j}^*$.

Simulates $\theta \sim \pi(\cdot)$ by simulatng $(\theta^*, S)|_{\alpha, H}$ then $\theta = \theta(\theta^*, S)$.

We henceforth work with the parameterisation (θ^*, S) .

The Chinese Restaurant Process, AKA CRP

The sequential simulation of parameters according to a $\Pi(\alpha, H)$ is analogous to restaurant seating!

- 1) There is $j = 1$ one customer in the restaurant. They are seated at table $k = 1$.
- 2) Suppose that after the customer $j = 2, 3, \dots, n$ arrives, there are n_k^j people seated at table k , and K_j tables in all are occupied.
- 3) The $j + 1$ 'st arrival chooses a new table wp $\alpha/(\alpha + j)$ and table k with probability $n_k^j/(\alpha + j)$.

This divides up n customers over K tables. The set S_k lists customers at table k .

Put an independent parameter $\theta_k^* \sim H$ on table $k = 1, 2, \dots, K$. Together, $\theta(\theta^*, S) \sim G$ with $G \sim \Pi(\alpha, H)$.

The CRP probability distribution

Let $S = (S_1, \dots, S_K)$ partition $\{1, 2, \dots, n\}$, and $n_k = |S_k|$. The CRP realises $S \in \Xi_{[n]}$ with probability

$$P_{\alpha, [n]}(S) = \frac{\Gamma(\alpha) \alpha^K \prod_{k=1}^K \Gamma(n_k)}{\Gamma(\alpha + n)}.$$

Intuition: Suppose the sequence of table assignments is

$$T = (1, 1, 2, 1, 2, 3, 3, 2, 2, 4)$$

for $n = 10$ customers so $S = \{\{1, 2, 4\}, \{3, 5, 8, 9\}, \{6, 7\}, \{10\}\}$.

Table assignment T and partition S are 1 to 1, so

$$\begin{aligned} P_{\alpha, [n]}(S) &= 1 \times \frac{1}{\alpha + 1} \times \frac{\alpha}{\alpha + 2} \times \frac{2}{\alpha + 3} \times \frac{1}{\alpha + 4} \times \frac{\alpha}{\alpha + 5} \\ &\quad \times \frac{1}{\alpha + 6} \times \frac{2}{\alpha + 7} \times \frac{3}{\alpha + 8} \times \frac{\alpha}{\alpha + 9} \\ &= \alpha^{K-1} 2! 3! 1! 0! \prod_{i=2}^{10} (\alpha + i - 1)^{-1}. \end{aligned}$$

Proof: The i 'th arrival brings a denominator factor $(\alpha + i - 1)^{-1}$, so the denominator is $\prod_{i=2}^n (\alpha + i - 1)$.

Now look at the numerator. Suppose the customers seated at table k are $S_k = \{i_1, i_2, \dots, i_{n_k}\}$. When i_1 arrived there was no-one sitting at table k , and $k - 1$ tables were occupied, so i_1 chose table k wp

$$\alpha / (\alpha + i_1 - 1).$$

After that, for $j = 2, \dots, n_k$, there were $j - 1$ seated at table k when i_j arrived, so i_j chose table k wp

$$(j - 1) / (\alpha + i_j - 1)$$

so the numerator factor for table k is $\alpha(n_k - 1)!$.

If we end up with K tables then there are $K - 1$ events in which a new table is chosen so

$$\begin{aligned} P_{\alpha, [n]}(S) &= \frac{\alpha^{K-1} \prod_{k=1}^K (n_k - 1)!}{\prod_{i=2}^n (\alpha + i - 1)} \\ &= \frac{\alpha^K \prod_{k=1}^K (n_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)} \end{aligned}$$

For n integer $\Gamma(n) = (n - 1)!$ and for $x > 0$, $x\Gamma(x) = \Gamma(x + 1)$ so we can write this in terms of Γ functions,

$$P_{\alpha, [n]}(S) = \frac{\Gamma(\alpha) \alpha^K \prod_{k=1}^K \Gamma(n_k)}{\Gamma(\alpha + n)} \quad [\text{EOP}]$$

Notice the RHS only depends on S through n_k , $k \in [K]$ so the CRP is exchangeable over labels $1, \dots, n$. The probability to get S doesn't depend on the order in which customers arrive.

The joint distribution of θ^*, S

The θ^* -values are simulated independently from H . Because

$$\pi(d\theta^*, S) = \pi(S)\pi(d\theta^*|S),$$

we have

$$\pi(d\theta^*, S) = P_{\alpha, [n]}(S) \prod_{k=1}^{K(S)} H(d\theta_k^*)$$

with $K(S) = |S|$. Here S acts like a model index.

For example, if H is a distribution on $\Omega = R^p$ and

$$H(dx) = h(x)dx,$$

so H has density h in R^p , then

$$\pi(\theta^*, S) = P_{\alpha, [n]}(S) \prod_{k=1}^{K(S)} h(\theta_k^*).$$

The dimension of θ^* is $K \times p$.

Let Ξ_n be the set of all partitions of $\{1, \dots, n\}$.

The state space of $\pi(\theta^*, S)$ is

$$\Omega^* = \bigcup_{S \in \Xi_n} \Omega^{K(S)} \times \{S\}.$$

If $A \subseteq \Omega^*$ and $(\theta^*, S) \sim \pi(\theta^*, S)$ above then

$$\Pr((\theta^*, S) \in A) =$$

$$\sum_{S \in \Xi_n} \int_{\Omega^{K(S)}} \mathbb{I}_{(\theta^*, S) \in A} \left[P_{\alpha, [n]}(S) \prod_{k=1}^{K(S)} h(\theta_k^*) \right] d\theta_1^*, \dots, d\theta_K^*$$

We evaluate this sort of thing using Monte Carlo!

Bayesian inference for a Dirichlet process mixture

Suppose our data are n samples $y_i \sim f(y_i|\theta_i), i = 1, \dots, n$ from a mixture with an unknown number of components.

In a DP model for a mixture $y_i \sim f(y_i|\theta_i)$ our prior for θ is

$$\theta \sim G \quad \text{with} \quad G \sim \Pi(\alpha, H).$$

In S, θ^* notation, Bayes rule for conditional probabilities is*

$$\pi(\theta^*, S|y) \propto f(y|\theta^*, S)\pi(S)\pi(\theta^*|S)$$

for $(\theta^*, S) \in \Omega^*$ so

$$\pi(\theta^*, S|y) \propto f(y|\theta^*, S)P_\alpha(S) \prod_{k=1}^K h(\theta_k^*).$$

The number of components K in θ^* is a random variable.

*This is a Very Useful Relation. It takes us straight to an expression for the posterior in a general DP-process mixture! We can just write this down.

Normal mixture for the Galaxy data

Each component of the mixture has an unknown mean and variance, $\theta_k^* = (\mu_k^*, \sigma_k^{*2})$. Our base distribution H has density

$$h(\theta_k^*) = h_\mu(\mu_k^*)h_\sigma(\sigma_k^{*2}).$$

If $S = (S_1, \dots, S_K)$ is a partition of $\{1, 2, \dots, n\}$ with $n = 82$, and $i \in S_{k_i}$ then

$$y_i | S, \mu^*, \sigma^* \sim N(\mu_{k_i}^*, \sigma_{k_i}^{*2}).$$

This determines the likelihood. Our “centring” DP-priors are

$$h_\mu(\mu_k^*) = N(\mu_k^*; \mu_0, \sigma_0^2)$$

and

$$h_\sigma(\sigma_k^{*2}) = \text{IG}(\sigma_k^{*2}; \alpha_0, \beta_0).$$

with $\mu_0 = 20, \sigma_0 = 10, \alpha_0 = 2$ and $\beta_0 = 1/9$ fixed hyper-parameters and I take $\alpha = 1$ in the DP-prior.

From our “Very Useful Relation”, the posterior is

$$\begin{aligned}\pi(S, \mu^*, \sigma^* | y) &\propto f(y | \mu^*, \sigma^*, S) \pi(\mu^*, \sigma^* | S) P_\alpha(S) \\ &\propto \prod_{k=1}^K \prod_{i \in S_k} N(y_i; \mu_k^*, \sigma_k^{*2}) \\ &\quad \times \prod_{k=1}^K N(\mu_k^*; \mu_0, \sigma_0^2) \text{IG}(\sigma_k^{*2}; \alpha_0, \beta_0) \\ &\quad \times \alpha^K \prod_{k=1}^K \Gamma(n_k).\end{aligned}$$

We dropped the denominator in the expression for $P(S)$ as it does not depend on S .

No $K!$ here. Sets $S_1, \dots, S_K$ are distinguished by their contents. The order of the S_k -indices is fixed via $\min(S_k) < \min(S_{k+1})$.

Remarks on conjugate priors and “Collapsed Gibbs”

I chose Normal/inv-Gamma for the μ^*/σ^* -prior to keep things simple and conjugate, so I could Gibbs-sample μ^*, σ^* .

If we just did straightforward MH-MCMC on μ^* and σ^* that wouldn't be necessary.

Conjugate priors are popular in this field as it allows us to integrate out $\theta^* = (\mu^*, \sigma^*)$ completely and just sample the discrete dbn $\pi(S|y)$. This is the “collapsed Gibbs sampler”. It is efficient. If our purpose is clustering, S is all we need anyway.

The downside is we cant model μ^* and σ^* with freedom.

Remarks on the choice of α

I chose $\alpha = 1$. This controls the prior distribution on the number of clusters. I used simulation (of the CRP) to check this distribution was sensible. The prior mean (PS4) is

$$E(K) = \sum_{i=1}^n \frac{\alpha}{\alpha + i - 1}.$$

which is about $E(K) = 5$ here.

This is a bit lower (was 10) than we used in our prior in L13 - we would take α a bit larger if we wanted to match that.

It is often straightforward to impose a hyper-prior on α and infer it along with everything else.

Gibbs sampler for μ^*, σ^* Iterate through the parameters sampling them conditionally. The conditional distribution for μ_k^* is

$$\pi(\mu_k^* | \mu_{-k}^*, \sigma^*, S, y) \propto N(\mu_k^*; \mu_0, \sigma_0^2) \prod_{i \in S_k} N(y_i; \mu_k^*, \sigma_k^{*2}).$$

Complete the square and find $\mu_k^* | \sigma_k^*, S_k, y \sim N(a, b)$ with

$$a = b \left(\frac{n_k \bar{y}_k}{\sigma_k^{*2}} + \frac{\mu_0}{\sigma_0^2} \right), \quad b = \left(\frac{n_k}{\sigma_k^{*2}} + \frac{1}{\sigma_0^2} \right)^{-1},$$

where $n_k = |S_k|$ and $\bar{y}_k = n_k^{-1} \sum_{i \in S_k} y_i$.

A similar calculation gives $\sigma_k^{*2} | \mu_k^*, S_k, y \sim \text{IG}(c, d)$ with

$$c = \alpha_0 + n_k/2, \quad d = \beta_0 + \frac{1}{2} \sum_{i \in S_k} (y_i - \mu_k^*)^2.$$